

Assignment 4 - Practice Quiz B: explanations

1. Answer: 4

Use prefix states. A match extends the diagonal; otherwise take the maximum of the two smaller prefixes.

2. Answer: LCS length of "aac" and "ab".

Indices are prefix lengths; subsequences need not be contiguous.

3. Answer: 4

B is a subsequence of A, so LCS length is $|B|$. All characters are distinct and B skips alternating characters, so the longest common substring has length 1.

4. Answer: 16

One optimal chronological plan $[l,k,r]$ is $[[0, 0, 1], [2, 2, 3], [0, 1, 3]]$. Its costs are $[5, 3, 8]$, totaling 16. Try each final split of each interval; add its sum once.

5. Answer: 43

Charge the first merged sum, then charge the sum of all three blocks.

6. Answer: 0

No merge occurs; $dp[i][i]=0$, regardless of the block size.

7. Answer: 17

One optimum selects vertices $[1, 3, 4]$, with weights $[7, 7, 3]$, totaling 17. Compute take/skip bottom-up; a selected parent forces its children to be skipped.

8. Answer: 19

When u is skipped, each child independently contributes $\max(\text{take}, \text{skip})$.

9. Answer: 25

Selecting u excludes each child; use their skip states.

10. Answer: 8

Every start from 0 to $n-m$ is a match, including overlaps.

11. Answer: 4

Compare proper prefixes with suffixes, or follow prefix-function fallback links.

12. Answer: 5

Only character-equal windows match. Equal-hash collision windows must be rejected after verification.

13. Answer: 14

The longest proper border is the pattern with the final "ab" removed.

14. Answer: 1

Compare whole suffixes lexicographically; a shorter prefix sorts first.

15. Answer: 2

The maximum adjacent-suffix LCP gives the longest substring occurring at distinct starts.

16. Answer: 8

After a comparison, decrease h by one if positive; reset to zero at rank zero.

17. Answer: $O(n)$

Count rounds and work per round, or stored characters. Changing the sorting method changes the doubling bound; Kasai is linear once SA exists.

18. Answer: 8

Duplicate patterns receive identical counts; do not discard a dictionary position.

19. Answer: 1

A pattern can end at the current trie state or along its suffix links. Count suffix outputs too.

20. Answer: 7

Reset query state and counters. Counts do not accumulate across independent texts.